

PLUG-AND-PLAY TEMPORAL FOURIER EMBEDDING FOR ROBUST LONG-HORIZON TRAFFIC FLOW FORECASTING

Peng Wang, Hairui Sun, Kanghua Hui

Civil Aviation University of China

ABSTRACT

Long-horizon traffic flow forecasting is essential for smart-city applications such as logistics scheduling, strategic traffic control, and urban mobility management. Although deep learning methods have significantly advanced prediction performance, existing models often struggle to effectively capture long-range periodic patterns without increasing architectural complexity or computational cost. In this work, we present *Temporal Fourier Embedding* (TFE), a lightweight and model-agnostic feature augmentation that injects explicit periodic priors directly at the input level. By introducing learnable sine and cosine functions parameterized by dominant frequencies, TFE equips forecasting networks with prior knowledge of recurring cycles, enabling more accurate and robust learning of complex spatiotemporal dynamics. Extensive experiments on two real-world datasets demonstrate that TFE consistently improves both long-term forecasting accuracy and training efficiency across a range of backbone architectures. Due to its simplicity, flexibility, and plug-and-play design, TFE provides a practical and effective solution for enhancing long-horizon traffic prediction in intelligent transportation systems and related domains.

Index Terms— Traffic Flow Forecasting, Long-Horizon Prediction, Fourier Embedding

1. INTRODUCTION

Accurate traffic flow forecasting is essential for intelligent transportation systems and smart-city development, supporting everything from congestion management to urban planning. Long-horizon forecasting—predicting hours or days ahead—is particularly critical for applications such as heavy-vehicle scheduling and cross-regional logistics.

The proliferation of urban sensors has provided vast spatiotemporal data [1, 2], and recent advances in deep learning have substantially boosted predictive accuracy over traditional methods [3, 4]. Despite these advances, a critical challenge remains in long-horizon prediction: reliably capturing long-range temporal dynamics. As illustrated in Fig. 1,

traffic flow data exhibit pronounced periodic patterns (e.g., across multiple days). While such periodicity may be less pronounced in short-term intervals, it becomes increasingly dominant as the prediction horizon extends.

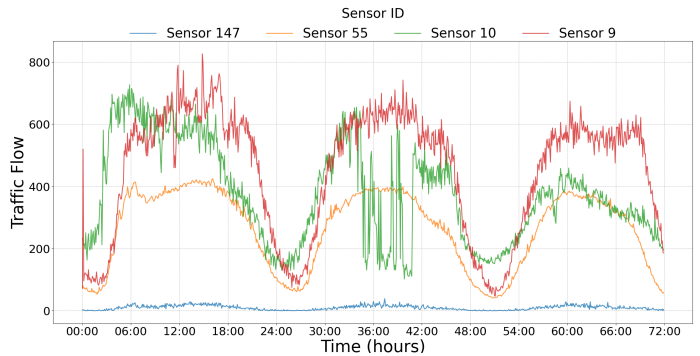


Fig. 1. Traffic flow time series from selected sensors in the PEMS08 dataset over three days. The data exhibits strong daily periodicity with distinct rush-hour peaks. The significant diversity in traffic volume and patterns across sensors underscores the complexity of the forecasting task.

However, existing approaches face significant limitations in modeling these patterns. Traditional methods for capturing periodicity, such as frequency decomposition or spectral analysis, often introduce considerable computational overhead, making them impractical for large-scale or real-time applications. On the other hand, many high-performing deep learning models, while effective at short-term forecasting, struggle to explicitly encode or leverage long-range periodic information, particularly when constrained by model complexity and efficiency requirements [5, 6].

This conflict underscores the need for a method that is both simple and effective—one that can seamlessly integrate key periodic signals into the forecasting process without imposing additional structural or computational burdens. To address this, we propose a powerful solution: *Temporal Fourier Embedding* (TFE). Our approach augments the model input with explicit periodic features, constructed via sine and cosine transformations that encode key temporal cycles (such as daily and weekly patterns) directly into the input space. TFE is lightweight, plug-and-play, and model-agnostic, al-

This work is supported by the Fundamental Research Funds for the Central Universities (No.3122023QD10) and the Major Program of the Natural Science Foundation of Tianjin (No.25JCLZJC00550).

lowing it to be seamlessly integrated into any existing traffic forecasting framework without modifying model architecture or incurring significant computational cost. By providing the model with clear periodic cues at the input level, TFE enables more efficient learning of long-range temporal dependencies. Extensive experiments demonstrate that our method not only enhances the model's ability to capture periodicity but also leads to faster training and superior performance across diverse predictive architectures.

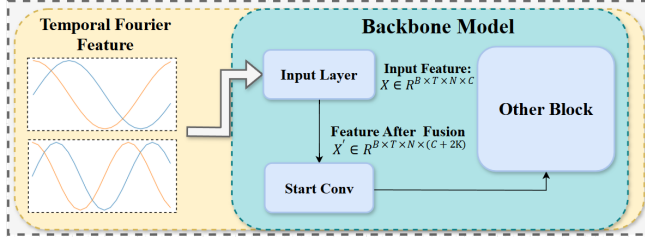


Fig. 2. The architecture of the Temporal Fourier Embedding (TFE) module. TFE generates periodic features (sine and cosine waves) representing temporal cycles and fuses them with the original input feature X . The resulting augmented feature X' , with an increased channel dimension, is then fed into the backbone model. This allows the model to capture long-range temporal dependencies more effectively without altering its core architecture.

2. RELATED WORK

Early approaches to traffic forecasting, including statistical models like ARIMA [7] and traditional machine learning, struggled to capture the complex, non-linear dynamics of traffic flow and often required laborious manual feature engineering. The advent of deep learning brought RNN-based models (e.g., LSTM) [4, 8] for temporal dependencies and CNN-based architectures like ST-ResNet [9] for spatial correlations. However, the reliance of CNNs on grid-based representations limited their direct applicability to real-world, irregular road networks. To overcome this limitation, Graph Neural Networks (GNNs) emerged as a natural fit for modeling complex road topologies. Foundational works like STGCN [10] integrated graph convolutions with sequence models. Subsequent innovations aimed to capture dynamic spatial relationships, such as the adaptive adjacency matrix in GWNet [11].

More recent models like DMSTGCN [12], SSTBAN [13], and the current state-of-the-art Extralonger [14], have further advanced the field by exploring dynamic graphs and enhancing long-horizon forecasting performance [15] [16]. Alongside GNNs, attention mechanisms have become another key component for enhancing spatiotemporal learning. For instance, ASTGCN [17] explicitly integrates spatial and temporal attention with GCNs. Recent trends also involve fusing

attention with multi-scale or diffusion-based methods to better capture long-range periodic patterns and complex structural dependencies [18, 19].

3. METHODOLOGY

3.1. Temporal Fourier Embedding (TFE)

To provide the forecasting model with explicit and adaptive knowledge of periodicity, we propose a Temporal Fourier Embedding (TFE) that augments the original input features with learnable periodic representations.

Given raw traffic data $\mathbf{X} \in R^{B \times C \times T \times N}$, where each feature vector at time step t includes speed, time-of-day, and day-of-week information, we introduce a set of learnable frequencies $\{f_k\}_{k=1}^K$ and compute corresponding sine and cosine embeddings for each time index:

$$\mathbf{f}_t = [\sin(2\pi f_1 t), \cos(2\pi f_1 t), \dots, \sin(2\pi f_K t), \cos(2\pi f_K t)], \quad (1)$$

where $t \in [0, T - 1]$ is the normalized relative time index within the lookback window, and K denotes the total number of frequencies. Each learnable frequency f_k is designed to capture a specific temporal cycle (e.g., $f_k = 1/24$ for daily periodicity).

These periodic features are concatenated with the original features to form an augmented representation:

$$\mathbf{x}_t^{\text{TFE}} = [\mathbf{x}_t, \mathbf{f}_t], \quad \mathbf{X}_{\text{TFE}} \in R^{B \times (C+2K) \times T \times N}. \quad (2)$$

By making the frequencies f_k trainable parameters, the model can automatically discover and align to dominant periodic patterns in the data (e.g., daily, weekly, or multi-scale cycles). This design injects prior knowledge of potential periodicity into the learning process, enabling the network to better capture long-range temporal dependencies, accelerate convergence, and improve long-horizon predictive performance.

4. THEORETICAL ANALYSIS

In this section, we provide a theoretical justification for the effectiveness TFE in two respects:

4.1. Why does TFE improve forecasting accuracy? (Q1)

Real traffic flow patterns y_t often decompose into a periodic component $p(t)$ plus a residual r_t , denoted as $y_t = p(t) + r_t$. We model the periodic component $p(t)$ as:

$$p(t) = \sum_{k=1}^K \left(a_k \sin(2\pi f_k t) + b_k \cos(2\pi f_k t) \right), \quad (3)$$

where t is the (normalized) time index. By explicitly including the basis functions in (3) as inputs, TFE reduces the approximation error to learning only the residual r_t , thereby decreasing model bias and improving long-horizon accuracy.

Table 1. Performance comparison on PEMS04, PEMS08 datasets. **Note:** Bold values indicate the best performance; underlined values denote the second best.

Dataset	Method	24 Time Steps			36 Time Steps			48 Time Steps		
		RMSE	MAE	MAPE	RMSE	MAE	MAPE	RMSE	MAE	MAPE
PEMS04	HA	81.57	56.47	45.49	106.58	76.01	68.84	127.28	93.37	94.62
	DCRNN	42.86	28.70	21.23	51.40	33.78	27.10	57.85	38.26	33.73
	GWNet	35.52	22.79	16.04	38.17	24.71	17.67	40.60	26.42	18.99
	GMAN	38.10	21.67	17.78	52.86	22.12	16.43	47.85	23.35	17.98
	AGCRN	34.44	21.63	14.65	38.19	24.15	16.33	38.26	24.18	16.31
	DMSTGCN	32.09	20.32	14.13	34.86	22.47	15.86	35.05	22.50	16.56
	DMSTGCN+TFE	31.76	19.88	13.81	34.09	22.04	15.66	34.91	22.31	16.32
	SSTBAN	33.15	20.36	14.73	34.15	20.82	14.83	35.81	21.66	15.90
	SSTBAN+TFE	32.85	19.81	14.21	33.88	20.51	14.31	35.37	21.26	15.11
	STPGNN	31.84	19.55	13.69	33.91	20.82	14.45	35.03	21.65	15.29
	STPGNN+TFE	31.28	19.27	<u>13.52</u>	33.57	<u>20.45</u>	14.24	34.54	21.37	15.14
	Extralonger	32.01	19.69	13.60	<u>33.52</u>	20.58	<u>14.13</u>	<u>34.41</u>	<u>21.15</u>	<u>14.65</u>
	Extralonger+TFE	<u>31.55</u>	<u>19.49</u>	13.45	33.05	20.37	13.73	33.75	20.75	14.16
PEMS08	HA	69.72	48.30	32.09	92.72	65.99	46.64	111.85	81.51	61.29
	DCRNN	33.34	22.60	15.46	39.37	25.82	18.53	45.64	30.47	25.10
	GWNet	29.47	19.07	12.25	33.54	21.76	13.68	34.20	22.60	14.16
	GMAN	34.29	17.38	15.66	35.89	17.21	16.33	48.54	18.70	16.81
	AGCRN	28.05	17.45	11.25	30.96	19.39	12.73	31.11	19.46	12.88
	DMSTGCN	26.55	16.75	11.44	28.50	18.15	12.64	28.94	18.34	12.93
	DMSTGCN+TFE	26.40	16.35	11.16	28.07	17.44	12.22	28.68	18.14	12.34
	SSTBAN	26.32	15.97	12.29	28.30	16.84	12.20	28.82	<u>16.94</u>	12.47
	SSTBAN+TFE	26.15	<u>15.82</u>	11.85	27.78	16.32	11.37	<u>28.66</u>	16.81	11.74
	STPGNN	<u>26.12</u>	16.05	<u>10.31</u>	28.01	16.99	11.61	28.95	17.62	11.90
	STPGNN+TFE	25.81	15.85	10.18	27.65	16.81	11.47	28.63	17.40	11.77
	Extralonger	26.39	15.89	10.49	28.10	16.72	<u>11.36</u>	29.01	17.31	<u>11.66</u>
	Extralonger+TFE	26.14	15.76	10.41	<u>27.71</u>	<u>16.62</u>	10.82	<u>28.66</u>	17.04	11.29

4.2. Why does TFE accelerate training convergence? (Q2)

Injecting periodic priors into the input lowers the effective complexity of the mapping the network must learn. Denote by Φ the original feature map and by Φ_{TFE} the TFE-augmented map. Then the empirical losses

$$\mathcal{L}(w) = \frac{1}{B} \sum_{i=1}^B \|\hat{y}_i(\Phi(w)) - y_i\|^2, \quad (4)$$

$$\mathcal{L}_{\text{TFE}}(w) = \frac{1}{B} \sum_{i=1}^B \|\hat{y}_i(\Phi_{\text{TFE}}(w)) - y_i\|^2 \quad (5)$$

exhibit a better-conditioned optimization landscape under standard smoothness assumptions. In particular, by offloading high-frequency periodic components to fixed basis inputs, TFE reduces the Lipschitz constant of $\nabla_w \mathcal{L}$ and the condition number of the Hessian, which in turn yields faster gradient-descent convergence. Empirically, this appears as steeper initial loss descent and fewer epochs to reach a given error threshold.

5. EXPERIMENTS

In this section, we conduct extensive experiments to address eight key questions: (Q1) How does TFE perform in long-term prediction? (Q2) How about extra-long-term forecast-

ing? (Q3) What is its impact on resource usage, including memory, inference, and training time? (Q4) How robust is TFE under noise and missing data?

5.1. Datasets, Baselines, and Experimental Setup

We evaluate the proposed **Temporal Fourier Embedding (TFE)** on two real-world traffic datasets: **PEMS04**, **PEMS08**.

TFE is integrated into a wide spectrum of forecasting models, including traditional statistical methods such as **HA** [20], classical deep learning models like **STGCN** [10], **ASTGCN** [21], and **DCRNN** [22], as well as more recent approaches such as **GWNet** [11], **GMAN** [23], **AGCRN** [24], **DMSTGCN** [12], **SSTBAN** [13], **STPGNN** [25], and the state-of-the-art **Extralonger** [14].

All experiments are conducted using PyTorch on an NVIDIA RTX 4090D GPU. TFE is applied at the input layer without modifying model architecture. We evaluate performance over long (24–48 steps) and extra-long (288–1440 steps) horizons using RMSE, MAE, and MAPE.

5.2. Experimental Result

5.2.1. Long-Term and Extra-Long-Term Forecasting Performance (Q1, Q2)

We evaluated the proposed TFE method on three state-of-the-art models—Extralonger, STPGNN, DMSTGCN, and SST-

Table 2. Performance comparison in extra-long-term scenarios. TS: Length of Time Steps.

Dataset	TS	Extralonger			Extralonger+TFE		
		RMSE	MAE	MAPE	RMSE	MAE	MAPE
PEMS04	288	40.46	24.18	16.70	39.65	23.46	16.10
	576	42.15	26.08	19.32	40.14	25.01	16.82
	864	42.98	26.60	19.36	41.99	25.11	16.94
	1152	42.98	26.88	19.33	42.23	25.12	16.93
	1440	43.00	26.62	19.24	42.11	24.90	16.51
PEMS08	288	33.76	19.38	13.05	32.75	19.18	12.63
	576	37.73	22.06	14.23	34.40	19.85	13.25
	864	38.57	22.57	15.04	35.36	20.35	13.13
	1152	40.45	23.50	16.03	37.16	20.25	13.38
	1440	41.44	23.73	17.90	37.55	20.60	14.02

BAN—using both the PEMS04 and PEMS08 datasets. As shown in Tables 1 and 2, integrating TFE with each backbone consistently improves forecasting performance across all evaluation metrics and prediction horizons, including both standard long-term and extra-long-term settings. Our statistical analysis further reveals that node complexity (temporal variance) follows a long-tail distribution with mean values of 16672.62 for PEMS04 and 12366.45 for PEMS08. TFE’s consistent gains across these heterogeneous sensors demonstrate its superior ability to stabilize predictions regardless of individual node volatility.

Notably, Extralonger+TFE achieves the best or second-best results in nearly every scenario. The improvements are especially pronounced under ultra-long-term horizons, where periodic patterns become increasingly dominant. These results demonstrate that explicit periodic cues provided by TFE enable models to better capture long-range cyclic trends in traffic data, leading to enhanced accuracy and robustness.

5.2.2. Resource Usage and Robustness (Q3, Q4)

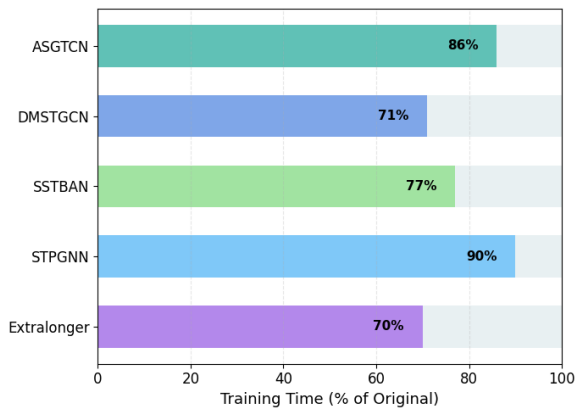


Fig. 3. Comparison of training time for different models with and without Temporal Fourier Embedding (TFE). Percentages indicate training time relative to each baseline.

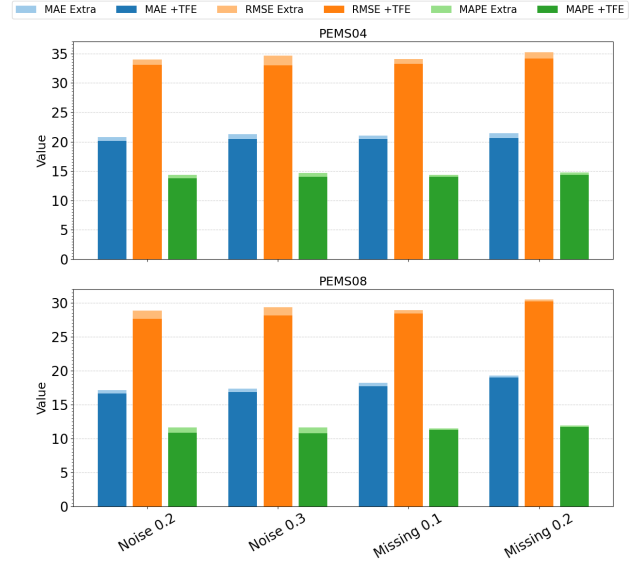


Fig. 4. Robustness evaluation of TFE under adverse conditions.

TFE is a plug-and-play input-level feature enhancement that introduces negligible overhead in memory and inference time, as it simply augments the model’s input without modifying network architecture. As shown in Fig. 3, integrating TFE leads to a notable reduction in training time across all evaluated models, highlighting its practical efficiency.

To further assess robustness, we evaluated TFE under challenging real-world scenarios, including moderate/severe Gaussian noise and missing data. As demonstrated in Fig. 4, TFE consistently improves predictive performance on both PEMS04 and PEMS08, even when the data is noisy or incomplete. These results confirm that TFE not only accelerates training, but also enhances the robustness and reliability of traffic forecasting models under adverse conditions.

6. CONCLUSION

This paper proposes a simple yet effective approach for long-horizon traffic flow forecasting by introducing Fourier-based periodic embeddings at the input level. Without changing model architectures or incurring additional computational cost, our method encodes key periodic patterns directly into the input, enabling any forecasting model to more easily capture long-term cycles. Experiments on real-world datasets show that this strategy consistently improves forecasting accuracy and accelerates model convergence across diverse architectures. Owing to its simplicity and generality, the proposed periodic embedding can be readily adopted in a wide range of spatiotemporal prediction tasks.

7. REFERENCES

- [1] Junbo Zhang, Yu Zheng, and Dekang Qi, "Deep spatio-temporal residual networks for citywide crowd flows prediction," in *Proceedings of the AAAI conference on artificial intelligence*, 2017, vol. 31.
- [2] Razvan-Gabriel Cirstea and Kieu, "Enhancenet: Plugin neural networks for enhancing correlated time series forecasting," IEEE, 2021.
- [3] Klaus Greff, Rupesh K Srivastava, Jan Koutník, Bas R Steunebrink, and Jürgen Schmidhuber, "Lstm: A search space odyssey," *IEEE transactions on neural networks and learning systems*, vol. 28, no. 10, pp. 2222–2232, 2016.
- [4] Yanjie Duan, LV Yisheng, and Fei-Yue Wang, "Travel time prediction with lstm neural network," in *2016 IEEE 19th international conference on intelligent transportation systems (ITSC)*. IEEE, 2016.
- [5] C. Park et al., "St-grat: A novel spatio-temporal graph attention networks for accurately forecasting dynamically changing road speed," in *Proc. WWW*, 2020.
- [6] Peng Wang, Longxi Feng, Yijie Zhu, and Haopeng Wu, "Hybrid spatial-temporal graph neural network for traffic forecasting," *Inf. Fusion*, vol. 118, pp. 102978, 2025.
- [7] Sangsoo Lee and Daniel B Fambro, "Application of subset autoregressive integrated moving average model for short-term freeway traffic volume forecasting," *Transportation research record*, vol. 1678, no. 1, 1999.
- [8] Rui Fu, Zuo Zhang, and Li Li, "Using lstm and gru neural network methods for traffic flow prediction," in *2016 31st (YAC)*. IEEE.
- [9] Junbo Zhang, Yu Zheng, and Dekang Qi, "Deep spatio-temporal residual networks for citywide crowd flows prediction," 2017, AAAI Press.
- [10] Bing Yu, Haoteng Yin, and Zhanxing Zhu, "Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting," 2018, ijcai.org.
- [11] Zonghan Wu, Shirui Pan, Guodong Long, Jing Jiang, and Chengqi Zhang, "Graph wavenet for deep spatial-temporal graph modeling," 2019, ijcai.org.
- [12] Liangzhe Han, Bowen Du, Leilei Sun, Yanjie Fu, Yisheng Lv, and Hui Xiong, "Dynamic and multi-faceted spatio-temporal deep learning for traffic speed forecasting," 2021.
- [13] Shengnan Guo and Lin, "Self-supervised spatial-temporal bottleneck attentive network for efficient long-term traffic forecasting," IEEE, 2023.
- [14] Zhiwei Zhang, Fandong Meng, Jie Zhou, Wenjuan Han, et al., "Extralonger: Toward a unified perspective of spatial-temporal factors for extra-long-term traffic forecasting," *arXiv preprint arXiv:2411.00844*.
- [15] J. Ma et al., "Less but more: Linear adaptive graph learning empowering spatiotemporal forecasting," *Proc. NeurIPS*, 2025.
- [16] J. Ma et al., "Bist: A lightweight and efficient bi-directional model for spatiotemporal prediction," *Proceedings of the VLDB Endowment*, vol. 18, no. 6, pp. 1663–1676, 2025.
- [17] Shengnan Guo, Youfang Lin, Ning Feng, Chao Song, and Huaiyu Wan, "Attention based spatial-temporal graph convolutional networks for traffic flow forecasting," 2019, AAAI Press.
- [18] Rui Xue, Shengjie Zhao, and Fengxia Han, "An embedding-driven multi-hop spatio-temporal attention network for traffic prediction," *IEEE Transactions on Intelligent Transportation Systems*.
- [19] Qingyong Zhang, Changwu Li, Fuwen Su, and Yuanzheng Li, "Spatiotemporal residual graph attention network for traffic flow forecasting," *IEEE Internet of Things Journal*, vol. 10, no. 13, pp. 11518–11532, 2023.
- [20] George EP Box, Gwilym M Jenkins, Gregory C Reinsel, and Greta M Ljung, *Time series analysis: forecasting and control*, John Wiley & Sons, 2015.
- [21] Shengnan Guo, Youfang Lin, Ning Feng, Chao Song, and Huaiyu Wan, "Attention based spatial-temporal graph convolutional networks for traffic flow forecasting," 2019, vol. 33.
- [22] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu, "Diffusion convolutional recurrent neural network: Data-driven traffic forecasting," 2018, OpenReview.net.
- [23] Chuanpan Zheng, Xiaoliang Fan, Cheng Wang, and Jianzhong Qi, "Gman: A graph multi-attention network for traffic prediction," in *Proceedings of the AAAI conference on artificial intelligence*, 2020, vol. 34, pp. 1234–1241.
- [24] Lei Bai, Lina Yao, Can Li, Xianzhi Wang, and Can Wang, "Adaptive graph convolutional recurrent network for traffic forecasting," *Advances in neural information processing systems*.
- [25] W. Kong et al., "Spatio-temporal pivotal graph neural networks for traffic flow forecasting," in *Proc. AAAI*, 2024.